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ML HW 3: Attention Mechanisms:

(1) a) Attention is a technique whereby the model can concentrate on the important words during the generation of every new word instead of considering all words equally. For example, in the translation task, as the model produces a particular word in English, it considers all the Telugu words but pays more attention to those that have high significance concerning the produced word. This becomes necessary due to the fact that every part of the sentence holds varying degrees of importance depending on the context, such that if a word like "strike" appears in the sentence, it would impact the interpretation of the previous phrases. The model decides relevance by comparing the current word's query to all other words' keys and produces scores that determine how much each token contributes. The scores are then converted into weights via softmax, and the weighted sum of the value vectors forms the output.

b) The query "asks" which tokens from the input sentence would be the most appropriate for generating the currently processed token in the output sequence. The projection of token representations to another space, referred to as query space, allows the model to learn how to extract features that define the connection between tokens.

c) The key of a token stands for what a word can provide to another word, i.e. how it acts like a description for something that the other word must pay attention to. For instance, using the translation analogy, each word in Telugu contains a key encoding info about itself, so that other words (or queries) can use it to see whether it is important or not. Dot product of keys and queries shows how good of a match they are: the larger it is, the better the token is at the current output position.

d) This value represents the actual info or content a token contributes once it has been deemed relevant by attention. Using a separate projection of values enables the model to transfer and retain the most valuable features needed for the output stage rather than only focusing on how relevant a token is.

e) They would most likely produce different attention patterns. Despite being the same word, each instance will appear in its own unique context, meaning that the model's query at each position will be different & will focus on different Telugu tokens. Attention is dependent on context, not just the word itself.

② a) $K^T = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$

$$S = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \rightarrow S \in \mathbb{R}^{2\times 2}$$

Entry (i,j) = dot product between key i and query j

Measures how much token j attends to token i

b) $d_k = 2 \rightarrow \sqrt{d_k} = \sqrt{2}$

$$\hat{S} = \frac{S}{\sqrt{2}} = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

c) $$A = \text{softmax}(\hat{S}) = \begin{pmatrix} \frac{1}{2} & \frac{1}{1+e^{1/\sqrt{2}}} \\ \frac{1}{2} & \frac{e^{1/\sqrt{2}}}{1+e^{1/\sqrt{2}}} \end{pmatrix} \approx \begin{pmatrix} 0.5000 & 0.3302 \\ 0.5000 & 0.6698 \end{pmatrix}$$

d) $$Z = VA = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 0.5000 & 0.3302 \\ 0.5000 & 0.6698 \end{pmatrix} = \begin{pmatrix} 1.0000 & 0.6604 \\ 1.5000 & 2.0094 \end{pmatrix}$$

e) First column of A is $\begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$. Query 1 attends equally to both tokens since both weights are 0.5.

③ a)

$$\text{Softmax}(s) = \begin{pmatrix} \frac{e^2}{e^2+e+1} \\ \frac{e}{e^2+e+1} \\ \frac{1}{e^2+e+1} \end{pmatrix}$$

all sum to 1 (just to verify)

b) $\text{softmax}(s') = \begin{pmatrix} \frac{e^4}{e^4+e^2+1} \\ \frac{e^2}{e^4+e^2+1} \\ \frac{1}{e^4+e^2+1} \end{pmatrix}$

Compared to a), the distribution becomes more peaked. The largest entry gets an even bigger weight, while the smaller entries get pushed closer to 0.

c) High raw scores for attention cause softmax to behave in a more extreme way, making attention highly focused on a single token. This could potentially lead to poor gradient flow, since once the softmax gets too peaked, the smaller entries get tiny gradients and the model becomes harder to train.

4 a) $X \in \mathbb{R}^{d_{model} \times n} = \mathbb{R}^{512 \times 6}$

$W^Q \in \mathbb{R}^{d_{model} \times d_k} = \mathbb{R}^{512 \times 64}$ Since $(W^Q)^T \in \mathbb{R}^{64 \times 512}$

$\boxed{Q = (W^Q)^T X \in \mathbb{R}^{64 \times 6}}$

b) $K \in \mathbb{R}^{64 \times 6} \rightarrow K^T \in \mathbb{R}^{6 \times 64}$

$Q \in \mathbb{R}^{64 \times 6}$ $\rightarrow$ $\boxed{S = K^T Q \in \mathbb{R}^{6 \times 6}}$

c) Applying softmax column-wise does not change the size, so $A \in \mathbb{R}^{6 \times 6}$

d) $V \in \mathbb{R}^{64 \times 6}$

$A \in \mathbb{R}^{6 \times 6}$ $\rightarrow Z = VA \in \mathbb{R}^{64 \times 6}$

The first dimension must equal d_v b/c the rows of V contain the value features and attention only mixes together across columns. It does not change the number of value features, so the output keeps first dimension $d_v = 64$.

5 a) For a single head, $W^Q, W^K, W^V \in \mathbb{R}^{d_{model} \times d_k} = \mathbb{R}^{256 \times 64}$

Each has $256 \times 64 = 16384$ parameters. Since there are 3 of them, it becomes 49152 parameters for one head. Across $h = 4$ heads: $49152 \times 4 = \boxed{196608}$

b) $W^O \in \mathbb{R}^{hd_v \times d_{model}}$; $hd_v = 4 \times 64 = 256$

$W^O \in \mathbb{R}^{256 \times 256} \rightarrow 256 \times 256 = 65536$

$\hookrightarrow 65536 + 196608 = \boxed{262144}$

c) $262144 \times 6 = \boxed{1572864}$

d) $W^Q, W^K, W^V \in \mathbb{R}^{512 \times 128}$

$512 \times 128 = 65536 \rightarrow \frac{512 \times 128}{256 \times 64} = 4 \rightarrow$ 4x bigger than before.

6 a) Multi-head attention is preferred for its ability to examine the input data from multiple angles simultaneously. With it, each head can analyze a unique aspect of the relation, while each individual head would have to mix that all into one representation. Multi-head attention is better at capturing more complex patterns without sacrificing the quality of information. This leads to overall more accurate representations than just using one attention head.

b) This indicates that the model is probably underutilizing the full capabilities

i. of its 8 heads b/c many of them have picked up on overlapping patterns rather than developing unique roles. It likely means that fewer heads could achieve similar performance, or that the model did not need as many distinct attention mechanisms for this task. It could mean a lack of efficiency but is not necessarily a problem.

ii. Attention dropout forces the model to not rely on any single hand, since some heads are randomly removed during training. This pushes the remaining heads to learn diff. patterns so the model can still perform well when others are missing. This reduces redundancy | increases diversity b/c as a result, heads are more likely to specialize.